



POWER BI DATA ANALYSIS

What is Business Intelligence?

Business intelligence (BI) is a combination of techniques and tools organizations use to analyze and present heaps of data purposefully.

Earlier

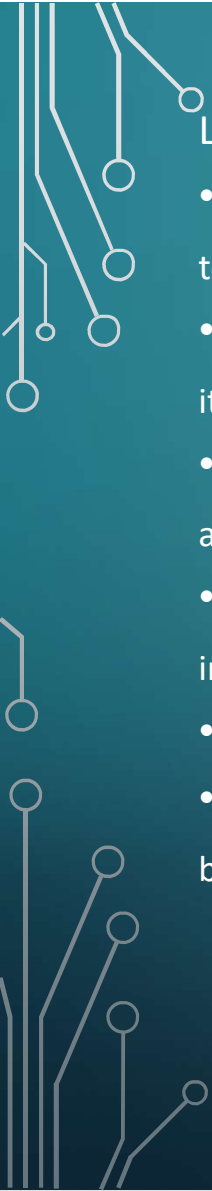
- Use of manual processes and spreadsheets for data analysis
- Time-consuming, error-prone, and inadequate data processing

Now

- Use of business intelligence tools to collect and analyze data
- Faster data processing to generate insights for informed decision-making

- 
- The slide features a dark teal background with a subtle gradient. White, stylized circuit lines with circular nodes are positioned along the left and right edges, creating a technical or digital aesthetic.
- By encompasses various technologies and methods such as data warehousing, data mining, predictive analytics, etc.
 - These technologies and methods allow you to use data effectively and efficiently to make data-driven decisions.
 - Various BI tools are available in the market, such as Power BI by Microsoft, Tableau by Salesforce, Zoho Analytics by Zoho and many others.

The selection of a tool over others is based on the requirements of a business.

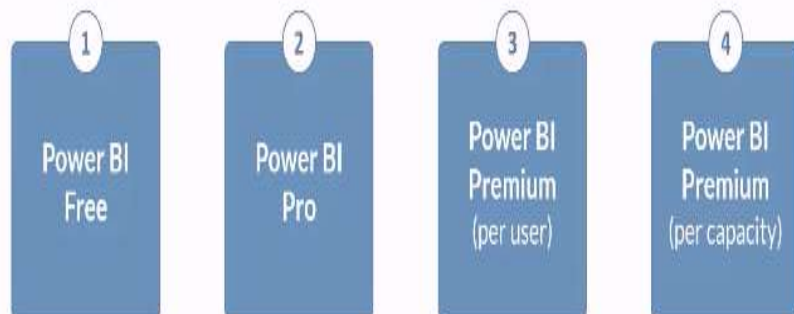


Let's learn about our BI and its components.

- Power BI is a cloud-based business intelligence platform provided by Microsoft that enables organizations to gather, transform, analyze and present data in a meaningful way.
 - It provides organizations with interactive visualizations and business intelligence capabilities, making it easy to create reports and dashboards.
 - Power BI provides organizations with data-driven insights that can inform their decision-making process and drive growth and success.
 - One of the salient features of Power BI is its ability to connect to a wide range of data sources, including on-premises and cloud-based data sources.
 - Organizations can access data from multiple sources and integrate it into a single view.
 - It provides a complete picture of the operations and allows organizations to make informed decisions based on data-driven insights.
- 

Power BI Licensing Types

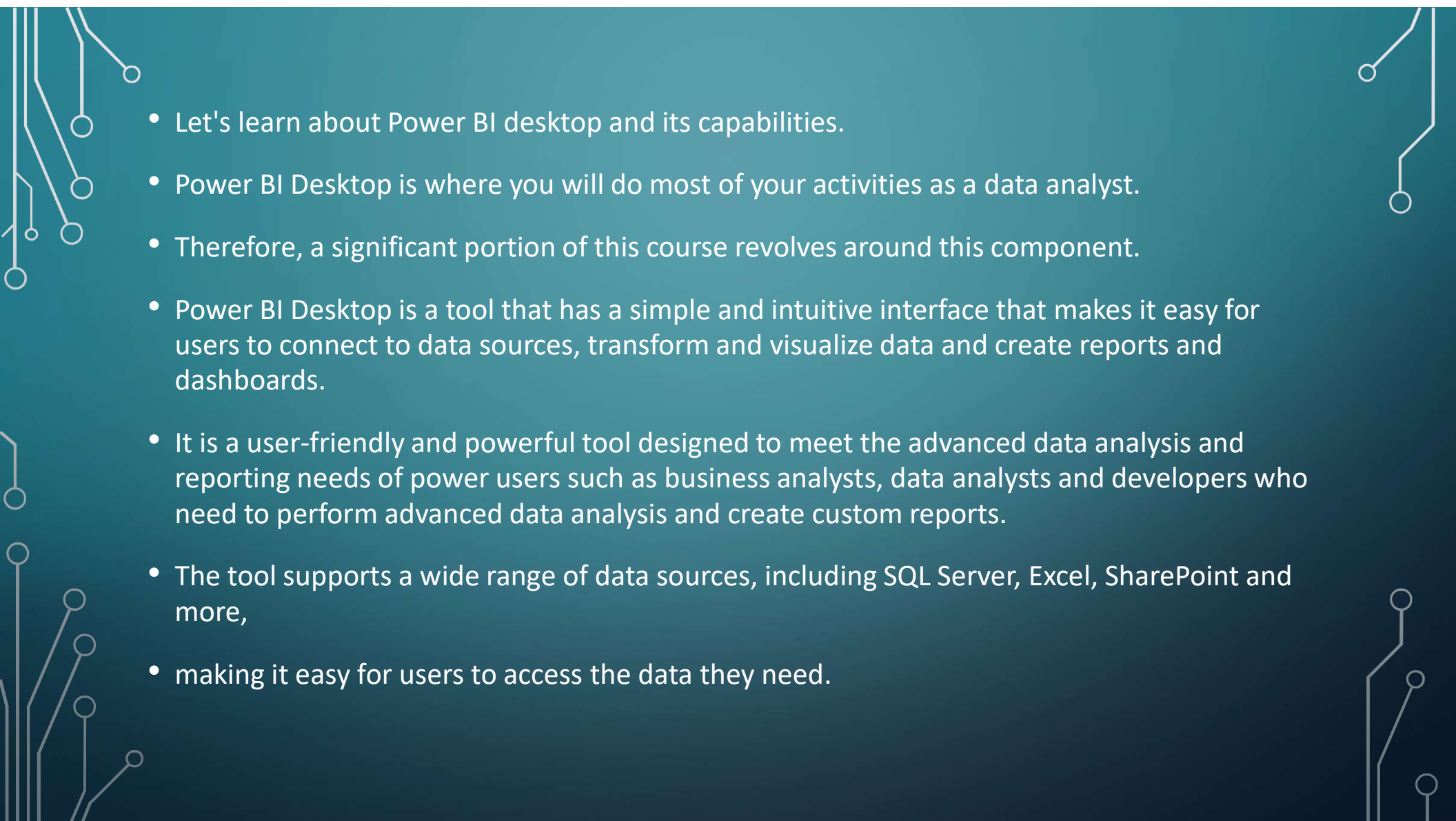
The four license types available for Power BI are:



Key Feature Differences

The key feature differences between the license types are:

Features	Power BI Free	Power BI Pro	Power BI Premium (per user)	Power BI Premium (per capacity)
Power BI Desktop Access	Available	Available	Available	Available
Collaboration and Sharing	-	Available	Available	-
Data Storage	10 GB	10 GB	100 TB	100 TB
Model Size Limit	1 GB	1 GB	100 GB	400 GB
Data Refresh Rate	1/Day	8/Day	48/Day	48/Day
App Integration	-	Available	Available	Available
On-premises Reporting	-	-	-	Available
Auto-scale	-	-	-	Available

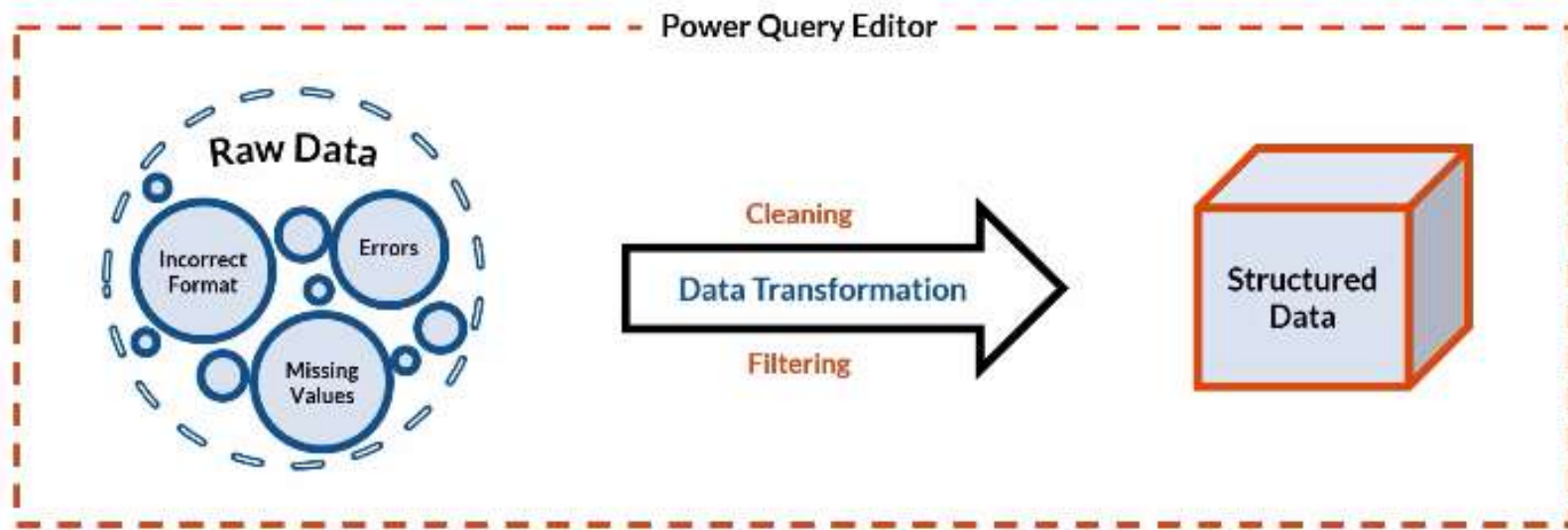
- 
- The background of the slide is a dark teal gradient. It is decorated with white, stylized circuit board traces. These traces enter from the top and bottom edges, featuring small circles at various points that resemble solder points or vias. The traces are more prominent on the left and right sides, framing the central text area.
- Let's learn about Power BI desktop and its capabilities.
 - Power BI Desktop is where you will do most of your activities as a data analyst.
 - Therefore, a significant portion of this course revolves around this component.
 - Power BI Desktop is a tool that has a simple and intuitive interface that makes it easy for users to connect to data sources, transform and visualize data and create reports and dashboards.
 - It is a user-friendly and powerful tool designed to meet the advanced data analysis and reporting needs of power users such as business analysts, data analysts and developers who need to perform advanced data analysis and create custom reports.
 - The tool supports a wide range of data sources, including SQL Server, Excel, SharePoint and more,
 - making it easy for users to access the data they need.

What is Data Source!!

- A data source is a location or application from which you extract, transform and load data as per the requirement.
- A data source can be a file, a database or an online service.
- Once data is extracted from its source, it is transformed to make it suitable for analysis.
- And then the transformed data is loaded into power BI.
- Power Query Editor, an ETL engine of Power BI, conducts this process.
- ETL stands for Extract, Transform and Load.
- You can connect to Excel Power BI datasets, data flows, Dataverse, SQL Server Analysis Services, Text or CSV Web and Odata feed.
- You can use the blank query option to write a query from scratch.

What is Data Transformation?

Data Transformation is the process of making the data suitable for analysis by performing various actions, such as cleaning and filtering.



Importance of Data Transformation

Data transformation is of great importance because it:

- Removes errors, inconsistencies, or missing values from the raw data
- Converts the raw data into a format that is suitable for analysis
- Prepares the raw data for data modeling



Data Transformation